

A Mechanism-Informed Gene Signature for Regenerative vs. Fibrotic Tendon Healing

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1. Introduction

1.1 The clinical problem

Tendon injuries heal by one of two routes, and the route taken decides whether a patient recovers function or stays impaired. In the regenerative route, resident progenitors rebuild the aligned type I collagen architecture that gives tendon its tensile strength. In the fibrotic route, the defect fills with disorganised scar. That scar restores continuity but not mechanical competence, leaving tissue that is stiffer, weaker, and prone to rupturing again.

The clinical burden is substantial. Rotator cuff tears are present in 20.7% of the general population (Yamamoto et al., 2010). A meta-analysis of surgical repairs reports re-tear in about 21% of cases at 12 to 24 months (Longo et al., 2021). Surgery restores anatomical continuity, but it cannot direct which healing program the tissue follows, and no drug currently shifts that balance. Cells make the decision early, well before any clinical assessment is possible.

1.2 Two progenitor populations decide the outcome

Harvey and colleagues resolved the cellular basis of this divergence. They showed that the tendon sheath houses two co-resident PDGFR α ⁺ progenitor populations with opposite fates (Harvey et al., 2019). Tubulin polymerisation-promoting protein family member 3-positive (Tppp3⁺) tendon stem and progenitor cells, or TSPCs, differentiate into tenocytes and regenerate tendon proper. Tppp3[−] fibro-adipogenic progenitors, called tendon FAPs or T-FAPs here, produce fibrotic matrix instead. Both express PDGFR α . Both sit in the same niche, and both are recruited by injury. Only Tppp3 separates them.

This architecture is not specific to tendon. The same fibro-adipogenic progenitor logic appears across tissues. FAPs govern the balance between regeneration and fibrosis in skeletal muscle (Scott et al., 2019), and Hic1-marked mesenchymal progenitors play a similar gatekeeping role in heart, skin, and brain (Soliman et al., 2020; Abbasi et al., 2020; Bernier et al., 2025). Any mechanism that decides which progenitor wins in tendon is therefore likely to generalise, which raises the value of mapping it carefully in a tractable system.

1.3 Innervation is an upstream control switch

Cherief and colleagues then showed that sensory innervation controls which of the two fates dominates after injury (Cherief et al., 2023). Their model is chemical-genetic, not surgical. Mice carry a TrkA^{F592A}

allele encoding an analogue-sensitive tropomyosin receptor kinase A, and the small molecule 1NM-PP1 selectively blocks TrkA kinase activity without cutting any nerves. Throughout this report, **“denervated” means this chemical-genetic TrkA blockade**, and “innervated” means the TrkA^{WT} control. No surgical denervation was performed in the source study or in this analysis.

Blocking TrkA/NGF signalling prevented TSPC expansion and impaired functional repair, pushing healing toward fibrosis. Innervation therefore acts upstream of the fate decision. TGFβ and PDGF signalling have separately been implicated in tendon progenitor behaviour and matrix remodelling (Kaji et al., 2020; Howell et al., 2017; Moser et al., 2021), which places the innervation signal in a wider niche-signalling context.

1.4 Gap 1 — the executing transcriptional program is unmapped

What remains unknown is the transcriptional program that converts innervation status into a fate decision inside the progenitor. Earlier work establishes *that* innervation controls the choice. Nothing maps the transcription-factor network that carries it out. Identifying that network is the difference between a phenomenological observation and a mechanism something can be aimed at.

The unit of analysis used here is the **regulon**: a transcription factor together with the target genes it is inferred to regulate, where each inferred link is kept only if the factor’s binding motif is present in the target’s regulatory region. A regulon’s activity can be scored in every individual cell. That makes it possible to ask which transcriptional programs, and not just which individual genes, separate two populations.

1.5 Gap 2 — no objective quantitative readout of healing

A second, more practical gap is that tendon healing is still assessed by eye. Human histological scoring remains the standard, and there is no reproducible quantitative alternative. Semi-quantitative scoring systems need trained assessors, vary between them, and compress a graded biological process into a handful of ordinal categories.

A score computed straight from expression data would be reproducible by construction, and it would resolve intermediate states that ordinal scoring cannot. The intent here is a research endpoint, a way to compare conditions, timepoints, and interventions on one scale. It is not a clinical diagnostic. The available datasets cannot support that stronger claim, and §4.4 explains why.

1.6 Objectives and hypothesis

This project addresses both gaps through two linked arms. A mechanistic arm maps the regulatory circuit. An applied arm compresses it into a validated quantitative score.

Objective 1 (mechanistic). Identify the transcription-factor regulons and upstream niche signals that govern the TSPC/T-FAP decision, by combining regulon inference (pySCENIC), fate trajectory modelling (CellRank 2), and ligand–receptor analysis (LIANA) across two independent mouse tendon datasets.

Objective 2 (applied). Compress the resulting gene programs into a minimal panel, targeted at roughly 10 to 30 genes, and a continuous Healing Index that scores any tendon single-cell dataset on a regenerative-to-fibrotic axis.

The link between the two arms is the intended novelty. Instead of selecting features from the whole transcriptome, the mechanistic arm supplies a biological prior that constrains the candidate space before any machine learning runs. The result is a mechanism-informed signature.

The original hypothesis was that innervated-niche signals activate a set of TSPC-biasing regulons, and that denervation removes those signals and lets T-FAP-driven fibrosis proceed. The data did not support it. The hypothesis is stated here in its original form; §4.1 addresses the discrepancy.

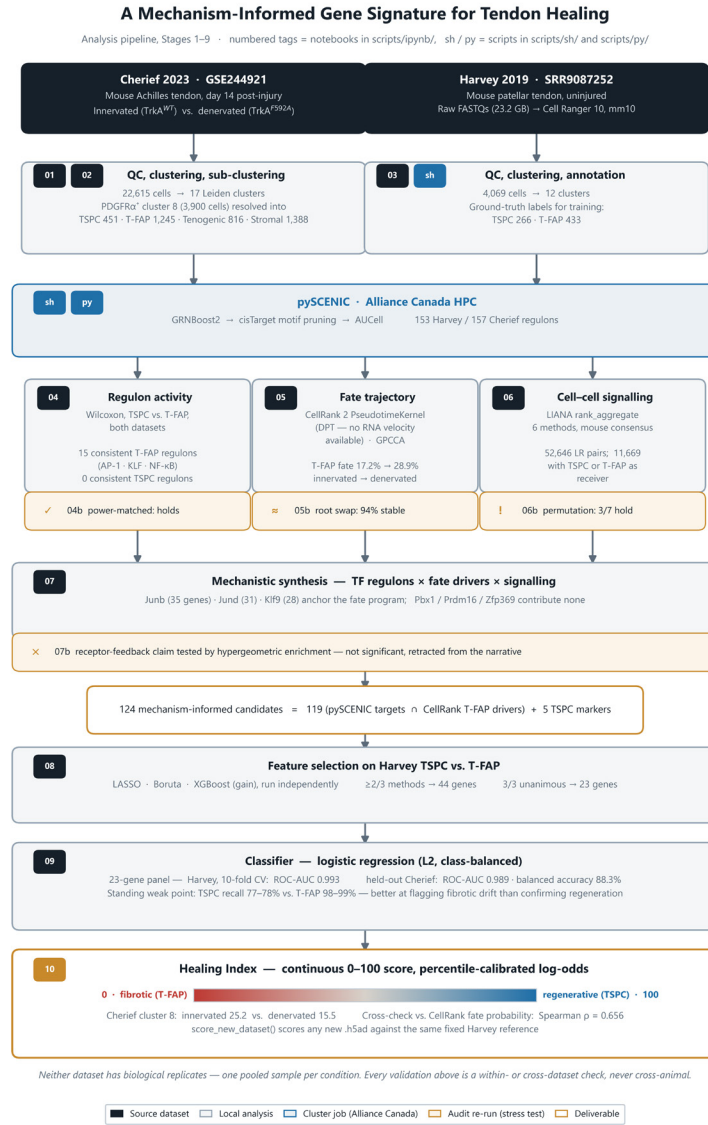


Figure 1. Analysis pipeline. Two published mouse tendon scRNA-seq datasets (black) enter a common preprocessing path, diverge through three independent mechanistic analyses (transcription-factor regulon inference, fate-trajectory modelling, and ligand–receptor signalling), and reconverge on a 124-gene mechanism-informed candidate set. That set is reduced to a 23-gene panel and a continuous Healing Index. Amber boxes mark re-analyses that stress-tested a headline claim: ✓ confirmed, ≈ partially robust, ! partially failed, ✗ retracted. Numbered tags correspond to notebooks in `scripts/ipynb/`. Neither dataset contains biological replicates, so every validation shown is a within- or cross-dataset check.

2. Methods

2.1 Datasets

Two published mouse tendon scRNA-seq datasets were used. Each supplies something the other cannot. Harvey 2019 provides uninjured tissue in which TSPC and T-FAP are separately characterised, so it supplies ground-truth training labels. Cherief 2023 provides the innervated versus denervated contrast the mechanistic question needs. Because its labels were derived independently, it also serves as a held-out test set for the classifier.

All analysis was carried out in Python. The principal libraries are Scanpy and anndata for single-cell handling, pySCENIC for regulon inference, CellRank 2 for fate modelling, LIANA for ligand–receptor analysis, and scikit-learn and XGBoost for classification, with numpy, scipy, and matplotlib underneath. Each is cited at the step where it is used, and all versions are pinned in `environments/` (§2.11).

Table 1. Datasets.

	Cherief 2023	Harvey 2019
Accession	GEO GSE244921	SRA PRJNA506218 / SRR9087252
Tissue	Mouse Achilles tendon, day 14 post-injury	Mouse patellar tendon, uninjured adult
Conditions	Innervated (TrkA ^{WT}) vs. denervated (TrkA ^{F592A} + 1NM-PP1)	Single homeostatic condition
Input	Processed count matrices	Raw FASTQ, 23.2 GB
Cells after QC	22,615	4,069 total; 266 TSPC and 433 T-FAP used for training
Role	Mechanistic analysis; held-out classifier test	Ground-truth labels; classifier training
Reference	Cherief et al. (2023)	Harvey et al. (2019)

2.2 Preprocessing and quality control

Both datasets went through the same Scanpy workflow (Wolf et al., 2018), but with deliberately different quality-control thresholds. One dataset is uninjured tissue and the other is day-14 post-injury, and that difference matters.

Harvey raw reads were aligned with Cell Ranger 10.0.0 against the `refdata-gex-mm10-2020-A` reference on the Digital Research Alliance of Canada (DRAC) infrastructure (Zheng et al., 2017). Cell Ranger called 8,484 cells. Filtering at a minimum of 700 detected genes, below 15% mitochondrial reads, and below 5,000 detected genes retained 4,069 cells, or 48.0%. Cherief matrices were filtered at a minimum of 200 genes, below 20% mitochondrial reads, and below 7,500 genes, which kept 22,615 of 25,545 cells, or 88.5%.

The asymmetry is deliberate. Harvey is uninjured patellar tendon with low ambient RNA, so a 700-gene floor removes empty and low-complexity droplets without discarding real cells. Cherief is day-14 injured Achilles tendon with heavy immune infiltrate, where many genuine cells, particularly small immune cells, fall below that floor. Applying Harvey’s thresholds to Cherief would have deleted a large part of the injury response the analysis is about.

Both mitochondrial cutoffs are looser than the 5 to 10% often used by default, and both were set from the data. Plotting total counts against mitochondrial percentage gives the familiar L-shaped distribution, where an upper-left arm of low-count, high-mitochondrial cells marks damaged cells whose cytoplasmic RNA has leaked away. The cutoffs were placed to remove that arm while leaving the main cloud intact, and the upper gene limits were set the same way, targeting the doublet tail above the saturation curve of genes against counts.

These thresholds are this project's own and were not transcribed from the source publications. Harvey and colleagues report roughly 2,491 cells, while the filtering used here keeps 4,069, or 1.63 times that number. The gap reflects a more permissive threshold choice, not a reproduction of their pipeline. Downstream analyses use only the confidently annotated TSPC and T-FAP subsets.

Counts were then library-size normalised, log1p-transformed, reduced to 2,000 highly variable genes, and projected by principal component analysis.

2.3 Clustering and annotation

Cell types were assigned by Leiden clustering (Traag et al., 2019) on a UMAP embedding (McInnes et al., 2018), followed by marker-based annotation. The focal PDGFR α ⁺ population was then sub-clustered.

In Cherief, Leiden clustering at resolution 0.5 produced 17 clusters. Cluster 8, with 3,900 cells, is PDGFR α ⁺ and is the focal population. It is the only cluster where TSPC and T-FAP stay mixed.

Resolving that mixture needed re-embedding, not finer clustering. The neighbour graph and UMAP were recomputed on the 3,900 cluster-8 cells alone, re-using the principal components from the full object. This removes the dominant between-cell-type variation and exposes structure inside the cluster. Leiden was then swept across resolutions 0.2, 0.4, 0.6, 0.8, and 1.0, giving 4, 6, 10, 12, and 17 sub-clusters. Resolution 0.2 was selected, being the lowest resolution at which TSPC and T-FAP separate. Higher resolutions fragment the compartment without adding anything interpretable. The four resulting populations are Stromal (1,388 cells), T-FAP (1,245), Tenogenic-progenitor (816), and TSPC (451).

Harvey clustering produced 12 clusters. TSPC was assigned to clusters 2 and 8 (Tppp3⁺, Prg4⁺; 266 cells) and T-FAP to clusters 3 and 11 (Tppp3⁻, Pdgfra⁺, Ly6a⁺/Pi16⁺; 433 cells). Both assignments were confirmed by Wilcoxon rank-sum differential expression against all other cells. Clusters without a confident marker-based assignment were labelled Unknown (1,025 cells) and dropped from all downstream analysis instead of being forced into a category.

One point matters for §3.7. **Cherief's TSPC and T-FAP labels are marker-based and were assigned independently of Harvey.** No label transfer was performed between the datasets. That is why the Cherief data can serve as a genuine held-out test and not a circular re-test of Harvey's own labels.

2.4 Transcription-factor network inference

Regulon activity was inferred independently in each dataset with pySCENIC (Aibar et al., 2017; Van de Sande et al., 2020). Cross-dataset consistency could then be used as the evidence standard, instead of a single pooled analysis.

The three-step pipeline ran on DRAC under SLURM. GRNBoost2 infers a co-expression network between transcription factors and candidate targets (Moerman et al., 2019). cisTarget then prunes that

network to links supported by the factor’s binding motif. AUCell scores the activity of each surviving regulon in each cell. The runs produced 153 regulons in Harvey and 157 in Cherief.

Differential activity was tested by Wilcoxon rank-sum on the AUCell scores, TSPC against T-FAP, separately within each dataset. A cross-dataset direction filter then kept only regulons that were differentially active in the same direction in both.

Inferring the networks separately and requiring agreement is a stronger design than pooling. Pooling would let a joint model absorb the differences between an uninjured patellar tendon and an injured Achilles tendon. Requiring independent replication turns those differences into a hurdle that any reported finding has to clear.

2.5 Fate trajectory modelling

Fate probabilities were estimated with CellRank 2 (Lange et al., 2022; Weiler et al., 2024) using a pseudotime kernel. The GEO deposit for Cherief 2023 lacks the spliced and unspliced counts an RNA-velocity kernel would need.

Diffusion pseudotime (Haghverdi et al., 2016) was computed with the root set to the Stromal cell of minimum first diffusion component. The biological prior behind that choice is that the Stromal sub-cluster, marked by the imprinted genes *Plagl1*, *Mest*, and *H19*, represents the least differentiated state. A generalised Perron cluster analysis estimator (Reuter et al., 2018) with `n_states=5` recovered all four annotated populations as distinct macrostates. Lineage driver genes were then defined as those whose expression correlates with a fate’s absorption probability, with significance assessed by Benjamini–Hochberg-corrected q-values (Benjamini & Hochberg, 1995), applied throughout via `scipy.stats.false_discovery_control` (Virtanen et al., 2020).

One limitation of the pseudotime belongs here and not in the Discussion. Diffusion pseudotime ranks Tenogenic-progenitor as the most root-like population, at mean 0.18, ahead of the Stromal root itself at 0.29. That ordering is biologically implausible given the imprinted-gene signature that motivated the root choice, so the trajectory ordering should not be treated as reliable. §2.10 describes how this was tested and §3.3 reports what it did to the results.

2.6 Cell–cell communication

Ligand–receptor signalling was compared between conditions with the LIANA rank-aggregate framework (Dimitrov et al., 2022), run separately on each condition instead of as one pooled model. The aggregate combines six methods: CellPhoneDB (Efremova et al., 2020), CellChat (Jin et al., 2021), Connectome (Raredon et al., 2019), NATMI (Hou et al., 2020), SingleCellSignalR (Cabello-Aguilar et al., 2020), and a log fold-change statistic, all run against the mouse consensus ligand–receptor resource.

After excluding small and uninterpretable cell groups, 20,047 of the 22,615 annotated cells remained: 11,478 innervated and 8,569 denervated. In total 52,646 ligand–receptor pairs were detected in both conditions, and 11,669 of those have TSPC or T-FAP as the receiving cell type. The differential metric is `delta_rank`, the difference between a pair’s aggregate rank in the denervated and innervated conditions.

The protocol reported here corrects the original run. The first analysis ran with permutation testing switched off, which left `delta_rank` carrying no significance test of any kind. The analysis was re-run

with 100 permutations to obtain CellPhoneDB-style p-values, and that corrected run is what this report uses. §3.4 gives what survived it.

2.7 Mechanistic integration

The three mechanistic analyses were integrated by intersecting their gene-level outputs. The union of the top 200 CellRank lineage-driver genes for each T-FAP macrostate came to 374 genes. Intersecting that with the 1,682 target genes of the cross-dataset consistent T-FAP regulons left 119 genes supported by both the trajectory model and the regulatory network. Those 119 were then re-aggregated by transcription factor, to separate the regulons that anchor the fate program from those that are merely differentially active.

2.8 Feature selection

The 119-gene mechanism-derived pool plus five established TSPC markers (Tppp3, Prg4, Igfbp6, Sema3c, and Cd34) gave 124 candidates. Three independent selection methods reduced that pool, chosen because they carry different inductive biases.

L1-penalised logistic regression (Tibshirani, 1996) kept 65 genes with non-zero coefficients at a regularisation strength of $C = 1.62$, chosen by 10-fold cross-validation. The Boruta algorithm (Kursa & Rudnicki, 2010), an all-relevant random-forest wrapper that tests each feature against randomised shadow copies of itself, as implemented in BorutaPy 0.4.3, confirmed 52 genes and left a further two tentative. XGBoost gradient boosting (Chen & Guestrin, 2016) kept 33 genes with above-mean gain importance. All three were implemented in scikit-learn (Pedregosa et al., 2011) and run independently on the same 124 candidates, using Harvey TSPC ($n = 266$) against T-FAP ($n = 433$).

Genes selected by at least two of the three methods form a 44-gene panel. Those selected by all three form a 23-gene core panel. Agreement across a penalised linear model, a permutation-based tree wrapper, and a boosted ensemble is stronger evidence than any single method's ranking, because the three make different assumptions about how expression relates to class.

2.9 Classifier and Healing Index

A logistic-regression classifier was trained on Harvey and evaluated on Cherief, then converted into a continuous score that stays comparable across datasets.

The classifier is L2-penalised and class-balanced, fit on standardised expression, and assessed by 10-fold stratified cross-validation within Harvey using out-of-fold predictions. The Harvey-fit scaler and classifier were then applied unmodified to Cherief as a held-out cross-dataset test. Logistic regression was chosen over a tree ensemble because the Healing Index needs well-calibrated continuous output more than it needs maximum discrimination.

The Healing Index required one design decision. The classifier's sigmoid probability is heavily saturated, with 65.0% of Cherief cluster-8 cells falling below 0.05 or above 0.95, since a model trained on two well-separated populations is confident almost everywhere. That squeezes most of the population onto two points and destroys resolution exactly where a continuous index is most useful. The index is therefore built on the raw decision-function log-odds, sign-inverted so higher values mean a more regenerative profile, then percentile-ranked against a fixed Harvey reference distribution. The fixed

reference keeps the 0 to 100 scale comparable across datasets; a score renormalized inside each new dataset would be an arbitrary within-sample ranking.

A documented function, `score_new_dataset()`, applies the same panel and reference to any new `.h5ad` file. It has not been demonstrated on data, because no third tendon dataset exists in this project.

2.10 Robustness and stress-testing protocol

Four headline findings were re-tested under stricter statistics before being reported. The protocols appear here so that the outcomes in §3 can be read against a pre-specified standard.

Table 2. Stress-testing protocol. Each test was specified before its outcome was known; the results appear in Table 5.

Finding	Test applied	Notebook
Zero consistent TSPC regulons	200-iteration stability selection; T-FAP downsampled to TSPC's n within each dataset, Wilcoxon re-run and BH-corrected per iteration, selection frequency ≥ 0.95 required in both datasets	04b
CellRank fate probabilities and the 119-gene candidate list	Full pipeline re-run under an alternate root (Tenogenic-progenitor, minimum DC1)	05b
LIANA differential pairs	100-permutation CellPhoneDB-style null	06b
Pdgfra receptor feedback	Per-dataset hypergeometric enrichment against each dataset's own GRNBoost2 gene universe, Fisher-combined, BH-corrected across the 15 regulons tested	07b

Stage 3B, label transfer from Harvey to Cherief, was planned and deliberately deferred. Its purpose was independent confirmation of the Cherief sub-cluster identities, and those were already supported by two lines of evidence: marker expression (§2.3) and regulon activity profiles (§3.2). The analysis is still available as a future validation step. Its absence does not affect the held-out status of the Cherief test set, because no transferred labels were used anywhere.

2.11 Computational reproducibility

All analysis is version-controlled and reproducible from three documented environments: Python 3.14.5 for the analysis, which produced every result reported here, Python 3.12 for the poster and figures, and Python 3.11.5 on DRAC for pySCENIC. All three are specified in `environments/`, with 23 pinned direct dependencies and a 161-package lock file for the analysis environment, along with the non-Python dependencies: Cell Ranger, the mm10 reference, and the cisTarget motif databases. Core numerical and plotting dependencies are numpy (Harris et al., 2020), scipy (Virtanen et al., 2020), matplotlib (Hunter, 2007), and anndata (Virshup et al., 2024). The lock file matters because it pins transitive dependencies that influence results without being imported by name; `pynndescent`, for example, builds the nearest-neighbour graph underlying UMAP, so its version can shift embedding coordinates.

One qualification belongs here. GRNBoost2 ran once without a fixed random seed, so the specific target-gene lists are not guaranteed to reproduce exactly, and conclusions resting on individual regulon-to-target edges should be read accordingly.

3. Results

3.1 A four-way split of the PDGFR α + compartment, shifting with innervation

Sub-clustering the PDGFR α + compartment resolved four populations. Two of them are the TSPC and T-FAP states of interest, and their relative abundance shifts with innervation status.

The 3,900 cluster-8 cells divide into Stromal (1,388), T-FAP (1,245), Tenogenic-progenitor (816), and TSPC (451). Each has a coherent marker profile. TSPC shows the highest Tppp3, at a scaled mean of 1.76, with Prg4 co-elevated and Sfrp2 essentially absent. T-FAP shows elevated Pi16 (1.63), Sfrp2 (1.35), and Pdgfra (1.48), together with a complement signature including C3 and C4b that the source literature does not describe. The Stromal population carries an imprinted-gene signature (Plagl1, Mest, H19), which is epigenetically distinct from the complement-driven T-FAP profile. The Tenogenic-progenitor population expresses tendon matrix genes (Thbs4, Kera, Col11a1), pointing to active commitment and not quiescence.

Composition shifts markedly with innervation. In the innervated condition (n = 2,094 cells), TSPC accounts for 15.8% of the compartment and T-FAP for 26.3%. In the denervated condition (n = 1,806), TSPC falls to 6.7% while T-FAP rises to 38.4%. That is a 2.4-fold depletion of TSPC and a 1.46-fold enrichment of T-FAP, both in the direction the innervation model predicts.

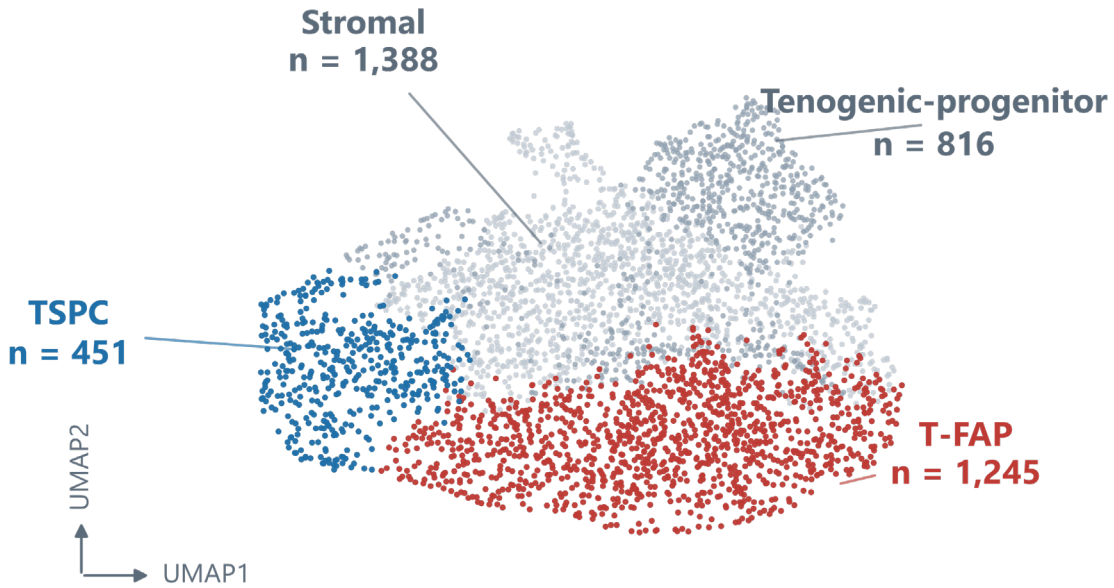


Figure 2. The PDGFR α + compartment resolves into four populations. UMAP of the 3,900 cells of Cherief 2023 cluster 8, re-embedded on those cells alone and sub-clustered with Leiden at resolution 0.2, chosen from a 0.2–1.0 sweep, then annotated by marker expression. TSPC (blue, Tppp3+/Prg4+, Sfrp2 near-absent) and T-FAP (red, Pi16+/Sfrp2+/Pdgfra+ with complement C3 and C4b) are the two fate endpoints. Stromal (imprinted signature Plagl1/Mest/H19) and Tenogenic-progenitor (Thbs4/Kera/Col11a1) are shown in grey. Both conditions are pooled here; the condition split is Figure 3.

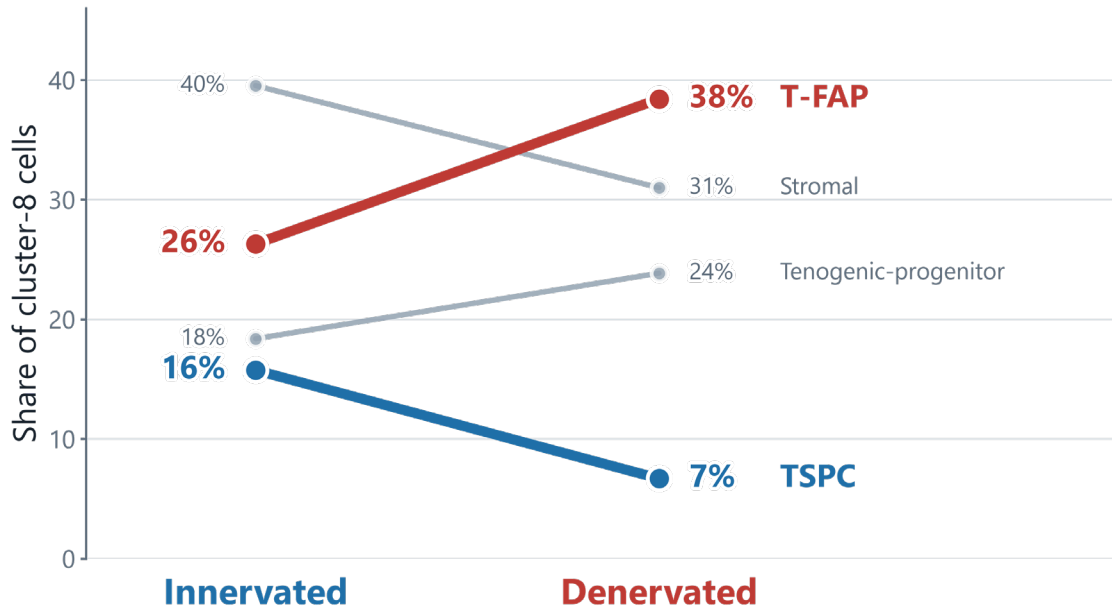


Figure 3. Denervation shifts compartment composition toward the fibrotic fate. Each population's share of the 3,900 cluster-8 cells, innervated (TrkA^{WT} , $n = 2,094$) against denervated ($\text{TrkA}^{\text{F592A}} + 1\text{NM-PP1}$, $n = 1,806$). TSPC falls from 15.8% to 6.7% while T-FAP rises from 26.3% to 38.4%. The two bystander populations move in opposite directions and are shown in grey for context. Percentages are shares of the compartment, not absolute cell counts.

3.2 Fifteen fibrotic regulons, zero regenerative ones

Cross-dataset regulon analysis found fifteen transcription-factor programs consistently associated with T-FAP identity in both datasets. It found none associated with TSPC.

The fifteen fall into coherent families: AP-1 (Fos, Fosb, Jun, Junb, Jund, Atf3), KLF (Klf6, Klf9), NF- κ B and inflammatory factors (Nfkb1, Irf1, Egr1, Cebpd), and three others (Pbx1, Prdm16, Zfp369). Together they make up a canonical activated-fibroblast program. Transcription factors associated with TSPC identity in Harvey, including Arnt2, Fli1, and E2f1, represent progenitor and cell-cycle programs. Those are reversed or absent in the injured Chierief tissue, which fits a difference between homeostatic and post-injury context.

The asymmetry invites an obvious objection, and it was tested. TSPC is the smaller population in both datasets, 266 against 433 in Harvey and 451 against 1,245 in Chierief, and Wilcoxon test statistics scale with sample size, so a rank-based cutoff structurally favours the larger group. The power-matched stability selection described in §2.10 reproduced 14 of the 15 T-FAP regulons, with Prdm16 dropping out and Zfx entering, both marginal, and again returned zero TSPC regulons. The three TSPC candidates raised during review fail for reasons unrelated to power: Cux1 and Mxd4 are not detected as regulons in Harvey at all, cisTarget having found no motif-supported target set for them, and Klf3 is genuinely null in Harvey at matched sample size, reaching significance in 5% of iterations.

This rules out a sample-size artefact. It does not rule out a real but small TSPC program that both datasets are jointly underpowered to detect, and §4.1 keeps that distinction explicit.

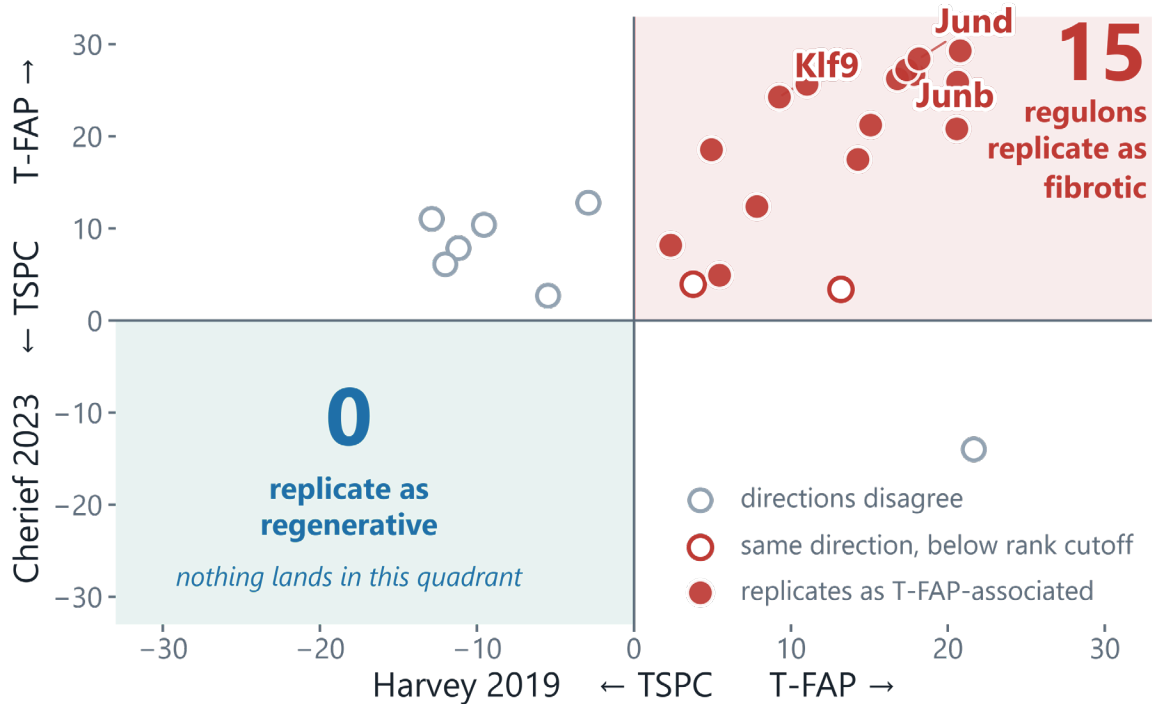


Figure 4. Fifteen regulons replicate as fibrotic; none replicate as regenerative. Each point is a transcription-factor regulon scored for differential AUCell activity between TSPC and T-FAP, independently in Harvey 2019 (x) and Cherief 2023 (y). Positive values on both axes mean T-FAP-associated activity in both datasets (upper right, shaded red). Negative values on both would mean a reproducible TSPC-associated program (lower left, shaded blue). Filled red points are the 15 regulons passing the cross-dataset direction filter; open red points share direction but fall below the rank cutoff; grey points disagree in direction between datasets. The lower-left quadrant is empty. Axes are signed Wilcoxon scores, and the quadrant boundaries sit at zero and not at a significance threshold.

3.3 Denervation shifts fate allocation toward T-FAP

Fate-probability modelling showed that denervation shifts the progenitor pool's allocation toward the fibrotic fate by roughly twelve percentage points.

Under the original Stromal root, mean TSPC fate probability falls from 0.826 in the innervated condition to 0.710 in the denervated condition, while aggregate T-FAP probability rises from 0.172 to 0.289. Under the alternate Tenogenic-progenitor root, the same values are 0.767 to 0.641 and 0.232 to 0.359.

These two results have different strengths and should be reported separately. The absolute probabilities are not root-invariant. The T-FAP baseline sits about 5 to 7 percentage points higher under the alternate root, so the absolute values are root-dependent point estimates and not measurements. The directional shift is robust, at roughly 12 percentage points toward T-FAP under either root.

The downstream result is more reassuring. Recomputing the candidate gene list under the alternate root reproduces 112 of the 119 genes, or 94%, with a Jaccard similarity of 0.89. Seven genes leave and seven enter. The gene list the applied arm consumes is therefore much more robust to the pseudotime problem than the population-level statistics are, which is what lets §3.6 proceed on it.

Top lineage drivers were, for TSPC: Igfbp6, Sema3c, Cd55, Crip1, Axl, Cd34, and Tppp3. For T-FAP they were Gdf10, Fmo2, Ccl11, Il6, and Steap4 in the first macrostate, and Cxcl14, Hp, Gas6, Angptl4, Sfrp1, C3, Sfrp2, and Hif1a in the second. Full ranked lists are in [data/cellrank/](#).

3.4 Niche signalling: what survives permutation testing

Of the ligand–receptor signals that differed most between conditions by rank, only a minority survived permutation testing. Only those are presented as findings.

Three pairs are supported at $p = 0.0$ in the condition where the signal was claimed to be strong. Bmp3 → Bmpr1b is an autocrine TSPC signal, consistent with an anti-fibrotic role. Pdgfc → Pdgfra is a stromal growth-factor signal arriving at TSPC and lost on denervation. Tnf → Notch1 is a macrophage-derived inflammatory signal arriving at TSPC and gained on denervation. Together they sketch a niche that supports progenitor maintenance when innervated and turns inflammatory when it is not.

Four previously prominent pairs are not supported. Efna1 → EphA3, Sema4a → Plxnd1, Wnt5a → Fzd4, and Sema3f → Nrp2_Plxna1 each return $p = 1.0$. Earlier project notes had described the first of these as the strongest differential signal overall, on the basis of its [delta_rank](#) alone. Across the top 20 pairs by [delta_rank](#) in each direction, only 8 of 20 innervated-favoured and 12 of 20 denervated-favoured pairs clear $p < 0.05$.

The rank-aggregate difference between two independently computed rankings is a hypothesis generator. On this dataset it identifies reproducible signalling differences only 40 to 60% of the time. Full tables are in [data/liana/](#).

3.5 An integrated circuit

Combining the three analyses gives a circuit in which innervation suppresses a fibrotic transcriptional program instead of activating a regenerative one.

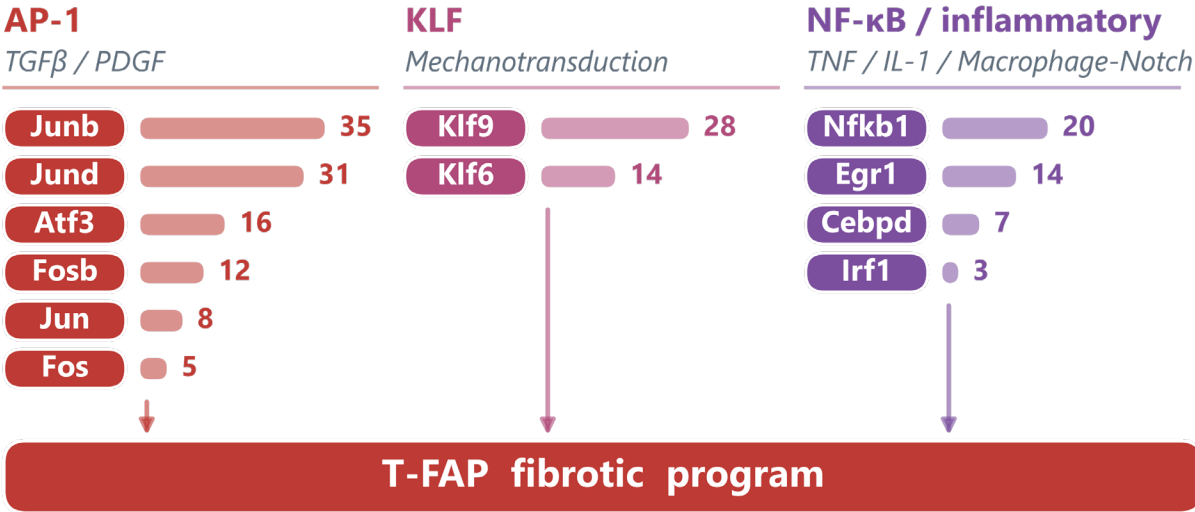
In the innervated state, PDGF-C and BMP3 signals are present, fate allocation favours TSPC, the AP-1/KLF/NF- κ B program stays quiet, and Tppp3+/Igfbp6+/Sema3c+ identity is maintained. Denervation loses PDGF-C, gains TNF → Notch1, activates those regulons, and expresses the T-FAP program, reinforced by Sfrp1 and Sfrp2 (Wnt antagonism), Hif1a, Il6, and C3.

Which regulons anchor the program is answerable from the data. Of the 119 mechanism-derived genes, Junb targets 35, Jun d 31, and Klf9 28, more than double any other regulon. Twelve of the fifteen contribute at least one gene. Pbx1, Prdm16, and Zfp369 contribute none, marking them as correlates of the T-FAP state and not drivers of the fate program.

One claim made during this analysis is retracted. An earlier synthesis reported that six T-FAP regulons target Pdgfra, and framed this as a feedback loop by which the fibrotic program reshapes the cell's own receptivity. That was a raw overlap count with no test against chance. Hypergeometric testing per dataset, Fisher-combined and BH-corrected across the fifteen regulons, finds nothing significant even at an uncorrected $p < 0.05$; the best case, Fos, reaches $p \approx 0.056$. The claim is withdrawn, and the circuit figure below has those edges removed.

Two caveats bound the rest of the circuit. The links from signal families to transcription-factor families, TNF to NF- κ B and TGF β to AP-1, are bridged from the literature and were not tested here. And cross-dataset agreement is much weaker for target genes than for regulon activity. Cebpd, Pbx1, Zfp369, and

Prdm16 share zero target genes between the two datasets, despite consistent differential activity in both. Individual transcription-factor-to-target edges should therefore be read as dataset-specific hypotheses and not established circuitry.



bar length = CellRank fate-driver genes contributed · *italic = upstream signal inferred from literature, not tested here*

Figure 5. The T-FAP transcriptional program, grouped by family and weighted by fate-driver contribution. The 12 regulons contributing at least one CellRank T-FAP lineage-driver gene, grouped into AP-1, KLF, and NF-κB/inflammatory families. Bar length and the adjacent number give the count of the 119 mechanism-derived candidate genes each regulon targets. Junb (35), Jund (31), and Klf9 (28) together anchor most of the program. Italicised text above each family is the upstream signal class inferred from the literature, **not tested in this work**. Pbx1, Prdm16, and Zfp369 are omitted: they are differentially active in both datasets but contribute zero fate-driver genes.

3.6 A 23-gene panel from three independent methods

Three feature-selection methods with different inductive biases converged on a 23-gene core panel.

Applied independently to the same 124 candidates, LASSO kept 65 genes, Boruta confirmed 52, and XGBoost kept 33. Forty-four genes were selected by at least two methods and 23 by all three. Of the 124 candidates, 41 were selected by none.

The 44-gene yield exceeds the 10 to 30 target set at the outset, and the reason is structural. The 124-candidate pool had already been filtered twice, first by regulon membership and then by lineage-driver correlation, before any selection method saw it. A high hit rate on a pre-enriched input is expected and does not indicate under-selective methods. The unanimous 23-gene panel falls inside the intended range and is carried forward as primary, with the 44-gene panel kept as a robustness check.

Two convergences were not designed in. First, all five hand-picked TSPC markers cleared the two-of-three bar, with Tppp3 and Sema3c selected unanimously, even though no method was told which candidates were TSPC-derived and which were T-FAP-derived. Second, 14 of the 23 core genes (61%) and 30 of the 44 (68%) are targets of Junb, Jund, or Klf9, the three regulons independently identified in §3.5 as anchoring the fate program. That annotation was applied after selection and never used inside it.

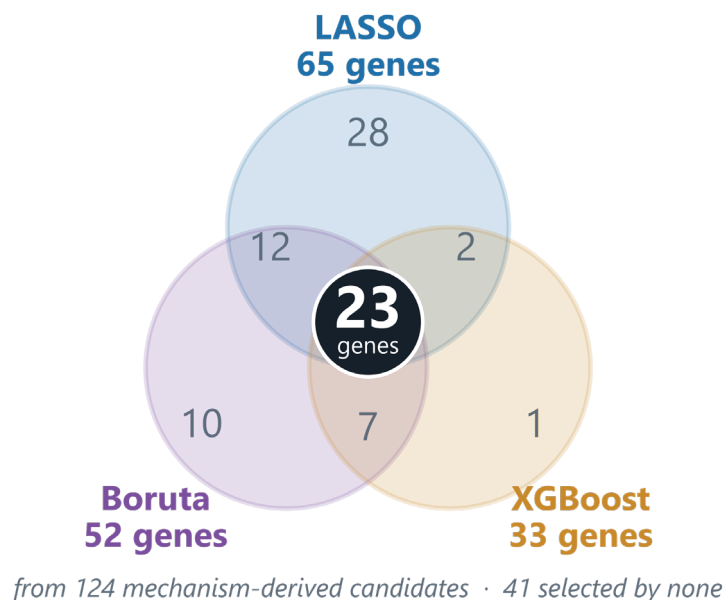


Figure 6. Three feature-selection methods converge on 23 genes. Overlap of the genes selected by LASSO (10-fold cross-validated L1 logistic regression, $C = 1.62$), Boruta (random-forest wrapper tested against randomised shadow features), and XGBoost (above-mean gain importance), each run independently on the same 124 mechanism-derived candidates using Harvey 2019 TSPC ($n = 266$) against T-FAP ($n = 433$). The 23 genes in the centre were selected unanimously and form the core panel. Adding the pairwise overlaps gives the 44-gene panel used as a robustness check. 41 of the 124 candidates were selected by no method.

Table 3. The 23-gene core panel. The regulon column lists the T-FAP regulons targeting each gene. TSPC markers were added as candidates and carry no regulon annotation.

Gene	Regulon(s) targeting it	Junb/Jund/Klf9 target	TSPC marker
Aebp1	Atf3		
C1ra	Cebpd		
C3	Junb	✓	
Camk2n1	Klf9	✓	
Cebpd	Atf3		
Cpxm1	Junb	✓	
Cxcl12	Jun		
Fth1	Jund	✓	
Igfbp7	Jund	✓	
Lpl	Cebpd		
Nfib	Junb, Klf9	✓	
Ptx3	Junb, Nfkb1, Klf9	✓	
Rps29	Junb	✓	
Scn7a	Jund	✓	
Sema3c	—		✓

Serping1	Junb	✓	
Sfrp2	Jun		
Slc38a2	Fosb, Junb	✓	
Slfn5	Junb	✓	
Steap4	Egr1, Junb	✓	
Tppp3	—		✓
Zfp36	Irf1, Jun, Fos, Atf3		
Zfp3611	Irf1, Jund, Fosb, Junb, Atf3, Klf9, Klf6	✓	

3.7 Cross-dataset generalisation

The Harvey-trained classifier generalised to Cherief without modification, and the smaller panel generalised at least as well as the larger one.

Table 4. Classifier performance on both panels. Harvey figures are out-of-fold predictions from 10-fold stratified cross-validation; Cherief figures come from applying the Harvey-fit model unmodified to an independently labelled dataset.

Panel	Harvey, 10-fold CV	Cherief, held out	Balanced accuracy
core_23	ROC-AUC 0.993 / 96.6% accuracy	ROC-AUC 0.989 / 93.5%	88.3%
full_44	ROC-AUC 0.995 / 97.7% accuracy	ROC-AUC 0.977 / 92.7%	88.2%

The 23-gene panel trails the 44-gene panel inside Harvey but matches or beats it on Cherief. That inversion is the expected signature of a larger panel carrying dataset-specific noise, and it is why the smaller panel goes forward.

Both panels share a real and consistent weakness. TSPC recall on Cherief is 77.4% for the 23-gene panel, meaning 102 of 451 true TSPC cells are called T-FAP. T-FAP recall is 99.3%, with only 9 of 1,245 cells misclassified. This asymmetry is why balanced accuracy, at 88.3%, sits well below raw accuracy at 93.5%. Two contributing factors are already established. The training data are imbalanced at 266 TSPC against 433 T-FAP, and §3.2 showed independently that TSPC carries the weaker transcriptional signature in this system. In practice the score identifies fibrotic drift reliably but cannot certify regenerative identity, and §3.8 inherits that asymmetry.

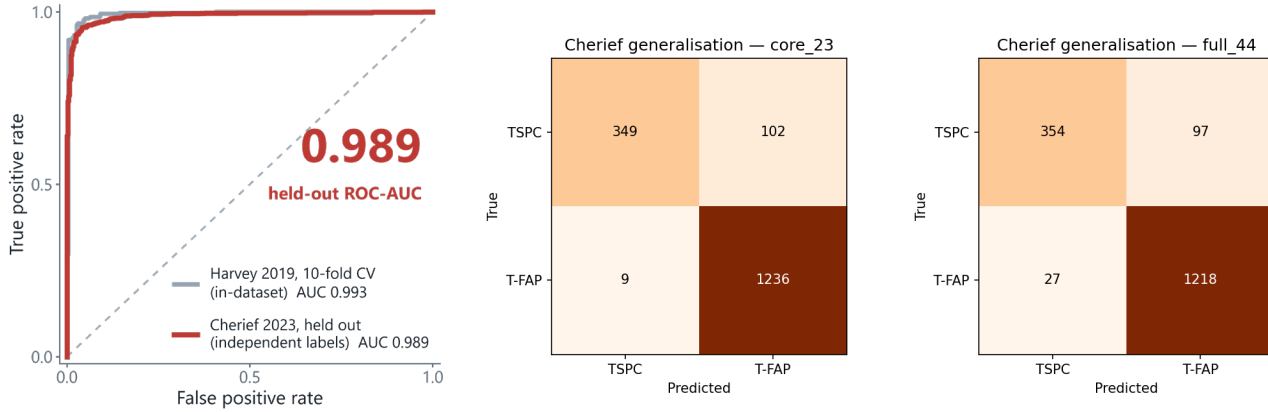


Figure 7. The 23-gene panel generalises across datasets. (a) ROC curves for the 23-gene classifier: grey, 10-fold stratified cross-validation within Harvey 2019 (in-dataset, AUC 0.993); red, the same Harvey-trained model applied unmodified to Cherief 2023 (held out, AUC 0.989). Cherief’s TSPC/T-FAP labels were assigned independently from marker expression and were never seen during feature selection or training. The dashed line is chance. (b) Confusion matrices on Cherief for both panels. The asymmetry is the panel’s main weakness and is visible directly: 102 of 451 true TSPC cells are called T-FAP, a recall of 77.4%, against only 9 of 1,245 T-FAP cells called TSPC, a recall of 99.3%.

3.8 A continuous Healing Index

Converting the classifier to a continuous score produced a 0 to 100 index on which higher values indicate a more regenerative profile. It separates conditions and places unseen cell types sensibly between the two fates.

Across all 3,900 Cherief cluster-8 cells, the mean Healing Index is 25.2 in the innervated condition, with a median of 18.9, and 15.5 in the denervated condition, with a median of 12.4. The direction is correct, and the distribution now has structure that the saturated probability of §2.9 could not show.

By population, the ordering is as expected at the extremes and informative in between. TSPC scores 52.0 and T-FAP 7.4, with Stromal at 22.3 and Tenogenic-progenitor at 21.2 between them. Neither intermediate population appeared in the training data in any form. That they land between the two trained classes indicates that the index measures a graded biological axis instead of a memorised decision rule.

An independent cross-check supports this. The Healing Index correlates with CellRank’s TSPC terminal-state fate probability at Spearman $\rho = 0.656$ across the same 3,900 cells, two estimates sharing no code path: one a 23-gene logistic model, the other a transcriptome-wide diffusion pseudotime model. The correlation is moderate, as expected. A dense band of cells sits at CellRank probability near 1.0 across a wide range of index values, which is the §2.5 pseudotime limitation made visible. Read this as directional corroboration, not validation against a ground truth that neither score has.

The population-level shift is partly compositional, since §3.1 established that T-FAP’s share of the compartment rises under denervation. A within-population signal survives that control: restricting to cells still labelled TSPC, mean $P(\text{T-FAP})$ rises from 0.217 to 0.381 after denervation. Cells that remain TSPC by marker identity have shifted toward the fibrotic profile. The change is graded drift, not a binary switch.

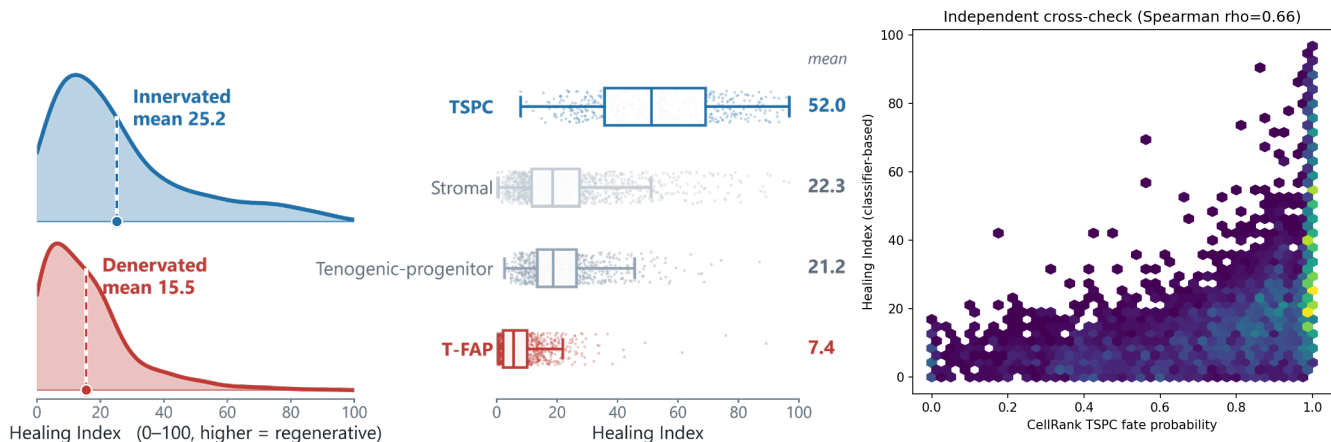


Figure 8. The Healing Index separates conditions and orders unseen populations sensibly. (a) Left, the distribution of the Healing Index across all 3,900 Cherief cluster-8 cells, split by innervation status (innervated mean 25.2, denervated 15.5; dashed lines mark the means; higher scores are more regenerative). Right, the same score by population. TSPC (52.0) and T-FAP (7.4) sit at the extremes as expected, while Stromal (22.3) and Tenogenic-progenitor (21.2), neither seen during training, fall between them and not at random. The score is percentile-calibrated log-odds against a fixed Harvey 2019 reference, so 0 to 100 stays comparable across datasets. **(b)** Independent cross-check against CellRank’s TSPC terminal-state fate probability, a method sharing no code path with the classifier. Spearman $\rho = 0.656$. The dense band at fate probability near 1.0 spanning a wide Healing Index range is the DPT pseudotime limitation from §2.5 made visible, and is why this is reported as directional corroboration.

4. Discussion

4.1 Innervation suppresses a fibrotic program rather than activating a regenerative one

The hypothesis that innervation activates a TSPC-biasing transcriptional program was not supported. The data show something different. A fibrotic program is the reproducible one, and innervation suppresses it.

That hypothesis (§1.6) predicted a set of regulons activated by innervated-niche signals and lost on denervation. Testing found zero transcription-factor regulons consistently associated with TSPC identity across the two datasets, against fifteen consistently associated with T-FAP. That result survived power-matched re-testing built specifically to detect whether it was an artefact of TSPC’s smaller sample size (§3.2). The specific candidate regulons proposed during review failed for reasons unconnected to power.

On this evidence fibrosis is the default program, and innervation holds it back. A pro-fibrotic AP-1/KLF/NF- κ B network is reproducibly detectable across an uninjured patellar tendon and an injured Achilles tendon, and no regenerative counterpart is. That is the pattern one would expect if the regenerative state is defined less by a distinctive transcriptional program than by the absence of an active fibrotic one.

The link to innervation requires care. The regulon comparison was TSPC against T-FAP, which is a comparison of cell types and not conditions. By itself it demonstrates nothing about innervation. Two separate results connect them. Denervation shifts the compartment toward T-FAP, in composition (§3.1)

and in modelled fate allocation (§3.3). And the Healing Index, which is oriented so that higher values indicate a more regenerative profile, falls from 25.2 in innervated tissue to 15.5 in denervated tissue (§3.8). Stated in the fibrotic direction, among cells still labelled TSPC the mean probability of the T-FAP class rises from 0.217 to 0.381 after denervation, a within-population shift that survives the compositional control. Suppression is the parsimonious reading of those two results together, though it is not a direct test. Demonstrating suppression would need a perturbation experiment, and this work does not include one.

One alternative remains. A real TSPC-biasing program may exist below the detection limit of two datasets holding 266 and 451 TSPC cells. What can be claimed is that no such program is detectable at this sample size. This was verified, not assumed.

4.2 Relation to prior literature

These results move the innervation-fate relationship from a phenomenological observation to a specific, testable transcriptional hypothesis.

Cherief and colleagues established *that* innervation controls the fate decision (Cherief et al., 2023). This work names *which* program is associated with the fibrotic outcome, an AP-1/KLF/NF- κ B network anchored by Junb, Jund, and Klf9, and shows that it replicates across two independent datasets from different tendons in different injury states.

AP-1 as a fibrotic effector fits the wider fibro-adipogenic progenitor literature, where analogous progenitor populations govern the regeneration-fibrosis balance in muscle, heart, skin, and brain (Scott et al., 2019; Soliman et al., 2020; Abbasi et al., 2020; Bernier et al., 2025). The finding is also compatible with the TGF β and PDGF signalling context established for tendon progenitors (Kaji et al., 2020; Harvey et al., 2019), though the specific signal-to-factor links remain literature-bridged and untested here.

Skeletal muscle supplies the closest precedent for the suppression model. Denervation there drives fibro-adipogenic progenitors to accumulate and to secrete IL-6 through persistent STAT3 activation, producing myofibre atrophy and fibrosis, and it does so without the macrophage infiltration and muscle stem cell activity that accompany regenerative repair (Madaro et al., 2018). Removing neural input is sufficient to move a progenitor compartment to a fibrotic output, with no injury needed to explain it. Regeneration in muscle depends on signals that innervation supplies, and fibrosis is what proceeds when those signals are absent. The tendon result reported here has the same shape, which places the inversion in §4.1 within an established cross-tissue pattern and supplies a reading of the zero-TSPC-regulon result: a permissive gate predicts no dedicated regenerative program to find, because what innervation contributes is restraint on the fibrotic one.

The parallel extends to a specific molecule. Il6 appears here as a CellRank T-FAP lineage driver (§3.3) and again in the integrated circuit (§3.5), placing the same cytokine on the fibrotic side in both tissues, and the one inflammatory niche signal that survived permutation testing, $Tnf \rightarrow Notch1$, is gained on denervation (§3.4). Two tissues and two unrelated analytical routes arrive at a neighbouring inflammatory axis. Whether the tendon and muscle mechanisms share more than a cytokine is untested here.

Sema3c is the strongest cross-analysis convergence in this work. It appears as a ligand in the innervated niche signalling analysis, as the second-ranked CellRank TSPC lineage driver, and as one of only two

markers selected unanimously by all three feature-selection methods. Three analyses with no shared code path converge on one gene. The LIANA component carries the permutation caveat from §3.4 and should not be overstated, but the convergence itself does not depend on that component. *Sema3c* is the most defensible single candidate for follow-up.

Its identity as a secreted class-3 semaphorin gives that convergence a possible mechanistic reading. Class-3 semaphorins signal through neuropilin co-receptors partnered with plexins, the receptor system that guides axons, and they act on the same NGF-responsive sensory neurons whose *TrkA* signalling the Chierief model blocks. NGF and *Sema3A* signalling interact directly in dorsal root ganglion growth cones, where NGF concentration sets sensitivity to semaphorin-induced growth cone collapse (Dontchev & Letourneau, 2002). The closer precedent is corneal: *SEMA3C* secreted by epithelium promotes sensory nerve regeneration and wound closure through *NRP2*, and hyperglycaemic loss of that signalling impairs both (Lee et al., 2019). A non-neuronal tissue cell using *Sema3C* to recruit sensory axons during repair is therefore an established mechanism in another tissue.

A TSPC-derived semaphorin would make the innervation axis reciprocal: sensory input maintains TSPC identity, and TSPCs help pattern the reinnervation that sustains them. Two observations here are consistent with that and neither establishes it. *Sema3c* is a TSPC lineage driver and a unanimous panel member, and the LIANA analysis independently surfaced *Sema3f* → *Nrp2_Plxna1* among the top differential pairs, reaching the same receptor system by a different route, though that pair failed permutation testing (§3.4). The limitation is structural: sensory axons in tendon have their cell bodies in the dorsal root ganglion, so dissociated tendon scRNA-seq captures no neuronal transcriptome, and no ligand-receptor claim with a neuronal target is testable in either dataset. This is a hypothesis for experimental work (§4.5).

4.3 What the stress-testing bought

Four headline findings were re-tested under stricter statistics. One held, one held only in part, and two were retracted.

Table 5. Stress-test register. Each test was specified in §2.10 before its outcome was known.

Finding	Test	Outcome
Zero consistent TSPC regulons	200-iteration power-matched stability selection	Confirmed. 14 of 15 T-FAP regulons replicate; TSPC remains zero
CellRank fate probabilities and the 119-gene candidate list	Full pipeline re-run under an alternate pseudotime root	Partly robust. Absolute probabilities shift 5–7 points; the directional shift and 94% of the gene list hold
<i>Efnal</i> → <i>Epha3</i> as strongest differential signal	100-permutation null	Retracted. $p = 1.0$
<i>Pdgfra</i> receptor feedback from T-FAP regulons	Per-dataset hypergeometric, Fisher-combined, BH-corrected	Retracted. Nothing significant even uncorrected

The methodological lesson generalises past this project. Both retracted findings came from procedures that produce compelling-looking results without testing them: a difference between two independently computed rank aggregates, and a raw set overlap between gene lists of unequal size. Both read as strong evidence and tested as noise. Rank differences and overlap counts are hypothesis generators, and treating them as evidence is a common error that is inexpensive to check for.

4.4 Limitations

Four limitations bound the interpretation, and the first matters most.

Neither dataset contains biological replicates. Each is a single pooled sample per condition, so every validation performed here, including cross-validation within Harvey, generalisation to Cherief, and the CellRank cross-check, is a within-dataset or cross-dataset comparison. None of it has been tested across independent animals. Cross-validation folds split cells within one pooled sample, estimating variability between cells and not between subjects, and a permutation test that shuffles cell labels inside one sample cannot address it either. This is a property of the available data and not of the analysis, and no re-analysis can resolve it.

Pseudotime ordering is unreliable. Diffusion pseudotime ranks Tenogenic-progenitor as more root-like than the Stromal population used as the root, which is biologically implausible. The absolute fate probabilities are therefore root-dependent (§3.3), and only the directional shift and the 94%-stable gene list should be relied on.

T-FAP is a project-defined annotation. Harvey's own authors excluded the equivalent cluster from their trajectory analysis, so it is not an independently established cell state in the source literature. This caveat propagates into every downstream stage, including the labels the classifier was trained on.

Cross-dataset technical differences are not corrected, beyond per-gene standardisation fit on Harvey alone. The two datasets come from different laboratories, different tendons (patellar against Achilles), and different biological states (uninjured against day-14 post-injury). That the classifier generalises anyway is reassuring, but it is not the same as having controlled for the differences, and the 93.5% held-out accuracy should be read as a real but not definitive estimate.

Two further limitations are recorded above and not repeated here: the asymmetric recall that makes the index better at detecting fibrotic drift than confirming regeneration (§3.7), and the literature-bridged signal-to-factor links in the circuit (§3.5).

4.5 Future directions

The immediate next step is the first test this signature has not yet faced.

Scoring newly generated laboratory tendon scRNA-seq would give the first biological-replicate test of the Healing Index. The scoring function and the fixed Harvey reference distribution already exist and need no further development, so the work is data collection.

A perturbation experiment would convert the suppression model from a parsimonious reading into a tested mechanism. Inhibiting AP-1 or Klf9 in the denervated model and asking whether the Healing Index recovers would test the causal claim directly, in a way no observational re-analysis can.

A second perturbation would test the reciprocal arm. Deleting Sema3c in TSPC and reading out sensory reinnervation alongside the Healing Index would establish whether progenitor-derived semaphorin signalling contributes to the innervation that maintains the population (§4.2).

Two smaller extensions follow. Stage 3B label transfer would independently confirm the Cherief sub-cluster identities. And spatial transcriptomics is the natural extension of the signalling analysis, since every ligand-receptor claim here is inferred from dissociated cells with no spatial information. The

inference that two cell types communicate currently rests on co-expression and not on proximity (Yue et al., 2023; Li et al., 2022).

4.6 Conclusion

A mechanism-informed 23-gene signature distinguishes regenerative from fibrotic tendon progenitors across two independent datasets, generalising out-of-dataset at ROC-AUC 0.989 without modification. The program that signature measures is fibrotic and not regenerative. Fifteen transcription-factor regulons replicate as T-FAP-associated across both datasets, and none replicate as TSPC-associated. The most useful reading is therefore an inversion of the original hypothesis. Fibrosis appears to be the default trajectory, and innervation the constraint that holds it back. The same relationship between denervation and fibrotic progenitor output is established in skeletal muscle (Madaro et al., 2018), which places the tendon result within a cross-tissue pattern. What remains untested is whether that constraint is causal, and whether any of this survives contact with biological replicates.

Data and code availability

All analysis code, intermediate results, and figures are in the project repository (<https://github.com/lifangy6/sunnybrook-tendon-fate-mapping>). Rather than reproduce them as appendices, the supporting tables are cited in place: the full LIANA differential tables with permutation p-values (`data/liana/`), CellRank lineage drivers for all three terminal states (`data/cellrank/`), the 44-gene panel and per-method selection flags for all 124 candidates (`data/feature_selection/`), and the three environment specifications with lock files (`environments/`). Notebooks are numbered by pipeline stage in `scripts/ipynb/`, and `scripts/README.md` maps each one to its inputs and outputs.

References

- Abbasi, S., Sinha, S., Labit, E., Rosin, N. L., Yoon, G., Rahmani, W., Jaffer, A., Sharma, N., Hagner, A., Shah, P., Arora, R., Yoon, J., Islam, A., Uchida, A., Chang, C. K., Stratton, J. A., Scott, R. W., Rossi, F. M. V., Underhill, T. M., & Biernaskie, J. (2020). Distinct regulatory programs control the latent regenerative potential of dermal fibroblasts during wound healing. *Cell Stem Cell*, 27(3), 396–412.e6. <https://doi.org/10.1016/j.stem.2020.07.008>
- Aibar, S., González-Blas, C. B., Moerman, T., Huynh-Thu, V. A., Imrichova, H., Hulselmans, G., Rambow, F., Marine, J.-C., Geurts, P., Aerts, J., van den Oord, J., Atak, Z. K., Wouters, J., & Aerts, S. (2017). SCENIC: Single-cell regulatory network inference and clustering. *Nature Methods*, 14(11), 1083–1086. <https://doi.org/10.1038/nmeth.4463>
- Benjamini, Y., & Hochberg, Y. (1995). Controlling the false discovery rate: A practical and powerful approach to multiple testing. *Journal of the Royal Statistical Society: Series B (Methodological)*, 57(1), 289–300. <https://doi.org/10.1111/j.2517-6161.1995.tb02031.x>
- Bernier, L.-P., Hefendehl, J. K., Scott, R. W., Tung, L. W., Lewis, C.-A., Soliman, H., Simm, S., Dissing-Olesen, L., Hofmann, J., Guo, D., DeMeglio, M., Rossi, F. M., Underhill, T. M., & MacVicar, B. A. (2025). Brain pericytes and perivascular fibroblasts are stromal progenitors with dual functions in cerebrovascular regeneration after stroke. *Nature Neuroscience*, 28(3), 517–535. <https://doi.org/10.1038/s41593-025-01872-y>
- Cabello-Aguilar, S., Alame, M., Kon-Sun-Tack, F., Fau, C., Lacroix, M., & Colinge, J. (2020). SingleCellSignalR: Inference of intercellular networks from single-cell transcriptomics. *Nucleic Acids Research*, 48(10), e55. <https://doi.org/10.1093/nar/gkaa183>
- Chen, T., & Guestrin, C. (2016). XGBoost: A scalable tree boosting system. In *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining* (pp. 785–794). Association for Computing Machinery. <https://doi.org/10.1145/2939672.2939785>
- Cherief, M., Xu, J., Li, Z., Tower, R. J., Ramesh, S., Qin, Q., Gomez-Salazar, M., Yea, J.-H., Lee, S., Negri, S., Xu, M., Price, T., Kendal, A. R., Fan, C.-M., Clemens, T. L., Levi, B., & James, A. W. (2023). TrkA-mediated sensory innervation of injured mouse tendon supports tendon sheath progenitor cell expansion and tendon repair. *Science Translational Medicine*, 15(727), eade4619. <https://doi.org/10.1126/scitranslmed.ade4619>
- Dimitrov, D., Türei, D., Garrido-Rodriguez, M., Burmedi, P. L., Nagai, J. S., Boys, C., Ramirez Flores, R. O., Kim, H., Szalai, B., Costa, I. G., Valdeolivas, A., Dugourd, A., & Saez-Rodriguez, J. (2022). Comparison of methods and resources for cell-cell communication inference from single-cell RNA-Seq data. *Nature Communications*, 13, 3224. <https://doi.org/10.1038/s41467-022-30755-0>
- Dontchev, V. D., & Letourneau, P. C. (2002). Nerve growth factor and semaphorin 3A signaling pathways interact in regulating sensory neuronal growth cone motility. *The Journal of Neuroscience*, 22(15), 6659–6669. <https://doi.org/10.1523/JNEUROSCI.22-15-06659.2002>

- Efremova, M., Vento-Tormo, M., Teichmann, S. A., & Vento-Tormo, R. (2020). CellPhoneDB: Inferring cell–cell communication from combined expression of multi-subunit ligand–receptor complexes. *Nature Protocols*, 15(4), 1484–1506. <https://doi.org/10.1038/s41596-020-0292-x>
- Haghverdi, L., Büttner, M., Wolf, F. A., Büttner, F., & Theis, F. J. (2016). Diffusion pseudotime robustly reconstructs lineage branching. *Nature Methods*, 13(10), 845–848. <https://doi.org/10.1038/nmeth.3971>
- Harris, C. R., Millman, K. J., van der Walt, S. J., Gommers, R., Virtanen, P., Cournapeau, D., Wieser, E., Taylor, J., Berg, S., Smith, N. J., Kern, R., Picus, M., Hoyer, S., van Kerkwijk, M. H., Brett, M., Haldane, A., del Río, J. F., Wiebe, M., Peterson, P., ... Oliphant, T. E. (2020). Array programming with NumPy. *Nature*, 585(7825), 357–362. <https://doi.org/10.1038/s41586-020-2649-2>
- Harvey, T., Flamenco, S., & Fan, C.-M. (2019). A Tppp3+Pdgfra+ tendon stem cell population contributes to regeneration and reveals a shared role for PDGF signalling in regeneration and fibrosis. *Nature Cell Biology*, 21(12), 1490–1503. <https://doi.org/10.1038/s41556-019-0417-z>
- Hou, R., Denisenko, E., Ong, H. T., Ramilowski, J. A., & Forrest, A. R. R. (2020). Predicting cell-to-cell communication networks using NATMI. *Nature Communications*, 11, 5011. <https://doi.org/10.1038/s41467-020-18873-z>
- Howell, K., Chien, C., Bell, R., Laudier, D., Tufa, S. F., Keene, D. R., Andarawis-Puri, N., & Huang, A. H. (2017). Novel model of tendon regeneration reveals distinct cell mechanisms underlying regenerative and fibrotic tendon healing. *Scientific Reports*, 7(1), 45238. <https://doi.org/10.1038/srep45238>
- Hunter, J. D. (2007). Matplotlib: A 2D graphics environment. *Computing in Science & Engineering*, 9(3), 90–95. <https://doi.org/10.1109/MCSE.2007.55>
- Jin, S., Guerrero-Juarez, C. F., Zhang, L., Chang, I., Ramos, R., Kuan, C.-H., Myung, P., Plikus, M. V., & Nie, Q. (2021). Inference and analysis of cell-cell communication using CellChat. *Nature Communications*, 12, 1088. <https://doi.org/10.1038/s41467-021-21246-9>
- Kaji, D. A., Howell, K. L., Balic, Z., Hubmacher, D., & Huang, A. H. (2020). Tgfb signaling is required for tenocyte recruitment and functional neonatal tendon regeneration. *eLife*, 9, e51779. <https://doi.org/10.7554/eLife.51779>
- Kursa, M. B., & Rudnicki, W. R. (2010). Feature selection with the Boruta package. *Journal of Statistical Software*, 36(11), 1–13. <https://doi.org/10.18637/jss.v036.i11>
- Lange, M., Bergen, V., Klein, M., Setty, M., Reuter, B., Bakhti, M., Lickert, H., Ansari, M., Schniering, J., Schiller, H. B., Pe'er, D., & Theis, F. J. (2022). CellRank for directed single-cell fate mapping. *Nature Methods*, 19(2), 159–170. <https://doi.org/10.1038/s41592-021-01346-6>
- Lee, P. S., Gao, N., Dike, M., Shkilnyy, O., Me, R., Zhang, Y., & Yu, F.-S. (2019). Opposing effects of neuropilin-1 and -2 on sensory nerve regeneration in wounded corneas: Role of Sema3C in ameliorating diabetic neurotrophic keratopathy. *Diabetes*, 68(4), 807–818. <https://doi.org/10.2337/db18-1172>

- Li, Y., Stanojevic, S., & Garmire, L. X. (2022). Emerging artificial intelligence applications in spatial transcriptomics analysis. *Computational and Structural Biotechnology Journal*, 20, 2895–2908. <https://doi.org/10.1016/j.csbj.2022.05.056>
- Longo, U. G., Carnevale, A., Piergentili, I., Berton, A., Candela, V., Schena, E., & Denaro, V. (2021). Retear rates after rotator cuff surgery: A systematic review and meta-analysis. *BMC Musculoskeletal Disorders*, 22, 749. <https://doi.org/10.1186/s12891-021-04634-6>
- Madaro, L., Passafaro, M., Sala, D., Etxaniz, U., Lugarini, F., Proietti, D., Alfonsi, M. V., Nicoletti, C., Gatto, S., De Bardi, M., Rojas-García, R., Giordani, L., Marinelli, S., Pagliarini, V., Sette, C., Sacco, A., & Puri, P. L. (2018). Denervation-activated STAT3–IL-6 signalling in fibro-adipogenic progenitors promotes myofibres atrophy and fibrosis. *Nature Cell Biology*, 20(8), 917–927. <https://doi.org/10.1038/s41556-018-0151-y>
- McInnes, L., Healy, J., & Melville, J. (2018). *UMAP: Uniform manifold approximation and projection for dimension reduction* (arXiv:1802.03426). arXiv. <https://doi.org/10.48550/arXiv.1802.03426>
- Moerman, T., Aibar Santos, S., Bravo González-Blas, C., Simm, J., Moreau, Y., Aerts, J., & Aerts, S. (2019). GRNBoost2 and Arboreto: Efficient and scalable inference of gene regulatory networks. *Bioinformatics*, 35(12), 2159–2161. <https://doi.org/10.1093/bioinformatics/bty916>
- Moser, H. L., Abraham, A. C., Howell, K., Laudier, D., Zumstein, M. A., Galatz, L. M., & Huang, A. H. (2021). Cell lineage tracing and functional assessment of supraspinatus tendon healing in an acute repair murine model. *Journal of Orthopaedic Research*, 39(8), 1789–1799. <https://doi.org/10.1002/jor.24769>
- Pedregosa, F., Varoquaux, G., Gramfort, A., Michel, V., Thirion, B., Grisel, O., Blondel, M., Prettenhofer, P., Weiss, R., Dubourg, V., Vanderplas, J., Passos, A., Cournapeau, D., Brucher, M., Perrot, M., & Duchesnay, É. (2011). Scikit-learn: Machine learning in Python. *Journal of Machine Learning Research*, 12, 2825–2830. <https://dl.acm.org/doi/10.5555/1953048.2078195>
- Raredon, M. S. B., Adams, T. S., Suhail, Y., Schupp, J. C., Poli, S., Neumark, N., Leiby, K. L., Greaney, A. M., Yuan, Y., Horien, C., Linderman, G., Engler, A. J., Boffa, D. J., Kluger, Y., Rosas, I. O., Levchenko, A., Kaminski, N., & Niklason, L. E. (2019). Single-cell connectomic analysis of adult mammalian lungs. *Science Advances*, 5(12), eaaw3851. <https://doi.org/10.1126/sciadv.aaw3851>
- Reuter, B., Weber, M., Fackeldey, K., Röblitz, S., & Garcia, M. E. (2018). Generalized Markov state modeling method for nonequilibrium biomolecular dynamics: Exemplified on amyloid β conformational dynamics driven by an oscillating electric field. *Journal of Chemical Theory and Computation*, 14(7), 3579–3594. <https://doi.org/10.1021/acs.jctc.8b00079>
- Scott, R. W., Arostegui, M., Schweitzer, R., Rossi, F. M. V., & Underhill, T. M. (2019). Hic1 defines quiescent mesenchymal progenitor subpopulations with distinct functions and fates in skeletal muscle regeneration. *Cell Stem Cell*, 25(6), 797–813.e9. <https://doi.org/10.1016/j.stem.2019.11.004>
- Soliman, H., Paylor, B., Scott, R. W., Lemos, D. R., Chang, C., Arostegui, M., Low, M., Lee, C., Fiore, D., Braghetta, P., Pospichalova, V., Barkauskas, C. E., Korinek, V., Rampazzo, A., MacLeod, K., Underhill, T. M., & Rossi, F. M. V. (2020). Pathogenic potential of Hic1-expressing cardiac

stromal progenitors. *Cell Stem Cell*, 26(2), 205–220.e8. <https://doi.org/10.1016/j.stem.2019.12.008>

- Tibshirani, R. (1996). Regression shrinkage and selection via the lasso. *Journal of the Royal Statistical Society: Series B (Methodological)*, 58(1), 267–288. <https://doi.org/10.1111/j.2517-6161.1996.tb02080.x>
- Traag, V. A., Waltman, L., & van Eck, N. J. (2019). From Louvain to Leiden: Guaranteeing well-connected communities. *Scientific Reports*, 9(1), 5233. <https://doi.org/10.1038/s41598-019-41695-z>
- Van de Sande, B., Flerin, C., Davie, K., De Waegeneer, M., Hulselmans, G., Aibar, S., Seurinck, R., Saelens, W., Cannoodt, R., Rouchon, Q., Verbeiren, T., De Maeyer, D., Reumers, J., Saeys, Y., & Aerts, S. (2020). A scalable SCENIC workflow for single-cell gene regulatory network analysis. *Nature Protocols*, 15(7), 2247–2276. <https://doi.org/10.1038/s41596-020-0336-2>
- Virshup, I., Rybakov, S., Theis, F. J., Angerer, P., & Wolf, F. A. (2024). anndata: Access and store annotated data matrices. *Journal of Open Source Software*, 9(101), 4371. <https://doi.org/10.21105/joss.04371>
- Virtanen, P., Gommers, R., Oliphant, T. E., Haberland, M., Reddy, T., Cournapeau, D., Burovski, E., Peterson, P., Weckesser, W., Bright, J., van der Walt, S. J., Brett, M., Wilson, J., Millman, K. J., Mayorov, N., Nelson, A. R. J., Jones, E., Kern, R., Larson, E., ... van Mulbregt, P. (2020). SciPy 1.0: Fundamental algorithms for scientific computing in Python. *Nature Methods*, 17(3), 261–272. <https://doi.org/10.1038/s41592-019-0686-2>
- Weiler, P., Lange, M., Klein, M., Pe’er, D., & Theis, F. J. (2024). CellRank 2: Unified fate mapping in multiview single-cell data. *Nature Methods*, 21(7), 1196–1205. <https://doi.org/10.1038/s41592-024-02303-9>
- Wolf, F. A., Angerer, P., & Theis, F. J. (2018). SCANPY: Large-scale single-cell gene expression data analysis. *Genome Biology*, 19(1), 15. <https://doi.org/10.1186/s13059-017-1382-0>
- Yamamoto, A., Takagishi, K., Osawa, T., Yanagawa, T., Nakajima, D., Shitara, H., & Kobayashi, T. (2010). Prevalence and risk factors of a rotator cuff tear in the general population. *Journal of Shoulder and Elbow Surgery*, 19(1), 116–120. <https://doi.org/10.1016/j.jse.2009.04.006>
- Yue, L., Liu, F., Hu, J., Yang, P., Wang, Y., Dong, J., Shu, W., Huang, X., & Wang, S. (2023). A guidebook of spatial transcriptomic technologies, data resources and analysis approaches. *Computational and Structural Biotechnology Journal*, 21, 940–955. <https://doi.org/10.1016/j.csbj.2023.01.016>
- Zheng, G. X. Y., Terry, J. M., Belgrader, P., Ryvkin, P., Bent, Z. W., Wilson, R., Ziraldo, S. B., Wheeler, T. D., McDermott, G. P., Zhu, J., Gregory, M. T., Shuga, J., Montesclaros, L., Underwood, J. G., Masquelier, D. A., Nishimura, S. Y., Schnall-Levin, M., Wyatt, P. W., Hindson, C. M., ... Bielas, J. H. (2017). Massively parallel digital transcriptional profiling of single cells. *Nature Communications*, 8, 14049. <https://doi.org/10.1038/ncomms14049>